

Part One: Reading Comprehension

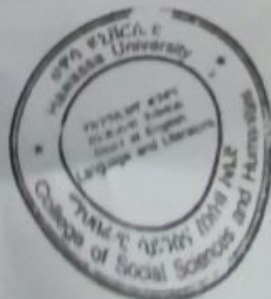
Instruction 1. Read the following passage carefully and answer the multiple choice items on the space provided (20 points).

(1) An **upsurge** of new research suggests that animals have a much higher level of **brainpower** than previously thought. If animals do have intelligence, how do scientists **measure** it? Before defining animals' intelligence, scientists defined what intelligence is not. Instinct is not intelligence. It is a skill programmed into an animal's brain by its **genetic heritage**. **Rate** conditioning is also not intelligence. Tricks can be learned by repetition, but no **real thinking** is involved. **Cuing**, in which animals learn to do or not to do certain things by following **outside** signals, does not demonstrate intelligence. Scientists believe that **insight**, the ability to use tools and communication using human language are all effective measures of the **mental ability** of animals.

(2) When judging animal intelligence, scientists look for insight, which they define as a **flash of** sudden understanding. When a young gorilla could not reach fruit from a tree, she noticed **crates** scattered about the lawn near the tree. She piled the crates into a pyramid, and then climbed on them to reach her reward. The gorilla's insight allowed her to solve a new problem without trial and error.

(3) The ability to use tools is also an important sign of intelligence. Crows use sticks to pry peanuts out of cracks. The crow exhibits intelligence by showing it has learned what a stick can do. Likewise, otters use rocks to crack open crab shells in order to get at the meat. In a series of complex moves, chimpanzees have been known to use sticks and stalks in order to get at a favorite snack - termites. To make and use a termite tool, a chimp first selects just the right stalk or twig. He trims and shapes the stick, then finds the entrance to a termite **mound**. While inserting the stick carefully into the entrance, the chimpanzee turns it skillfully to fit the inner tunnels. The chimp attracts the insects by shaking the twig. Then it pulls the tool out without scraping off any termites. Finally, he uses his lips to **skim** the termites into his mouth.

(4) Many animals have learned to communicate using human language. Some primates have learned hundreds of words in sign language. One chimp can recognize and correctly use more than 250 abstract symbols on a keyboard. These symbols represent human words. An amazing parrot can distinguish five objects of two different types. He can understand the difference between the number, color and kind of object. The ability to classify is a basic thinking skill. He seems to use language to express his needs and emotions. When ill and taken to the animal hospital for his first overnight stay, this parrot turned to go. "Come here!" he cried to a scientist who works with him. "I love you. I'm sorry. Wanna go back?"



(5) The research on animal intelligence raises important questions. If animals are smarter than once thought, would that change the way humans interact with them? Would humans stop hunting them for sport or survival? Would animals still be used for food, clothing or medical experimentation? Finding the answer to these tough questions makes a difficult puzzle even for a large-brained, problem-solving species like our own.

1. Crows use sticks to pry peanuts out of cracks. Which of the following is the kind of intelligence or conditioning the situation describes?

- A. rote learning
- B. tools
- C. communication
- D. instinct

2. The concluding paragraph of this passage infers that

- A. There is no definitive line between those animals with intelligence and those without.
- B. Animals are being given opportunities to display their intelligence.
- C. Research shows higher animal intelligence may fuel debate on ethics and cruelty.
- D. Animals are capable of untrained thought well beyond mere instinct.

3. According to the passage, which of the following is true about animals communicating through the use of human language?

- A. Parrots can imitate or repeat a sound.
- B. Dolphins click and whistle.
- C. Crows screech warnings to other crows.
- D. Chimpanzees and gorillas have been trained to use sign language or geometric shapes that stand for words.

4. In paragraph 3, what conclusion can be reached about the chimpanzee's ability to use a tool?

- A. It illustrates high intelligence because he is able to get his food and eat it.
- B. It illustrates instinct because he faced a difficult task and accomplished it.
- C. It illustrates high intelligence because he stored knowledge away and called it up at the right time.
- D. It illustrates high intelligence because termites are protein packed.

5. Which of the following is not a sign of animal intelligence?

- A. showing insight
- B. following cues
- C. using tools
- D. making plans

6. In paragraph 5, a species mentioned as "large-brained, problem-solving" is

- A. Chimpanzee
- B. Primate
- C. Parrot
- D. Human



7. According to the passage, what are the three effective measures of animal's intelligence?

- A. insight, instinct and cuing
- B. using tools, following signals, and communication using human language
- C. insight, the ability to use tools, and communication using human language
- D. all can be effective measures

8. According to the passage, to whom would finding the answer to questions mentioned in paragraph 5 be difficult?

- A. Scientists
- B. Human beings
- C. Animals
- D. Animals with intelligence

9. One of the animals is mentioned in the passage to exemplify the ability to use tools as a sign of intelligence.

- A. Crow
- B. Chimpanzee
- C. Parrot
- D. Termites

10. According to the passage, which one of the following is TRUE about rote conditioning?

- A. It is a skill programmed into an animal's brain by its genetic heritage.
- B. It is an intelligence which can be learned by repetition not involving thinking.
- C. It is an intelligence which can be learned by repetition involving thinking.
- D. It is a trick which can be learned by simple repetition.

11. The word **upsurge**, as it is used in paragraph 1 of the passage, most nearly means

- A. an increasingly large amount
- B. a decreasing amount
- C. a well-known amount
- D. an immeasurable amount

12. The word **crates**, as it is used in paragraph 2 of the passage, most nearly means

- A. wood
- B. container
- C. pole
- D. leather

13. The word **mound**, as it is used in paragraph 3 of the passage, most nearly means

- A. an increasingly large amount
- B. large pile of wet soil or mud
- C. large amount of any thing
- D. large amount of work

14. The word **skim**, as it is used in paragraph 3 of the passage, most nearly means

- A. to remove fat, cream, etc from the surface of a liquid
- B. to move quickly and lightly over a surface
- C. to read quickly in order to find the main point
- D. to eat quickly in order to get the best part

15. The word **puzzle**, as it is used in paragraph 5 of the passage, most nearly means

- A. riddle
- B. poem
- C. mystery
- D. poem

